

Get-Rescue — Distances & Times Calculated via Google Maps

Reconciled against the application, 9 October 2026. This is your edited document with the contents brought in line with what the system actually does. The section numbering, tables and field order are unchanged so the two can be read side by side. Yellow marks every place this version differs from yours, and every point that needs a decision from you.

Nine measurements use the Google Maps Distance Matrix API (real road distance and travel time, not straight-line estimates). Each is captured once, at a specific moment, and then held — nothing is tracked continuously. The nine are: the four start-to-breakdown figures at Accept, the two breakdown-to-drop figures at Picked-Up, and the three closure figures (`Vendor\_Distance`, `Vendor\_Round\_Trip\_Distance`, `Roundtrip\_Distance`). Google is deliberately not used anywhere else — per your 7 October instruction, every other distance falls back to straight-line.

Summary of changes from your version

- Locations B and E are captured and saved — your table marks both in red. Field names are given below so they can be checked directly in Zoho.
- Breakdown-to-Drop distance and ETA are built and live, not omitted. They are stored under different field names.
- The three changes you asked for — the `Roundtrip\_Distance` trigger, `Vendor\_Distance` branch 2 and `Vendor\_Round\_Trip\_Distance` branch 2 — were built on 8 October.
- Two of your section 5 points were already in place: the cancel-to-breakdown haversine change, and the 4-wheeler / 2-wheeler split.
- Four items need a decision from you before they can be built or changed. They are marked in place and listed again at the end.

Location data — locations to capture and save on the ticket

Both rows you marked red are in fact captured and saved, and have been since 11 September. The field names are given so they can be verified in Zoho directly.

SERVICE TYPE	VAR	LOCATION HEAD	SOURCE	==FIELD(S) ON CREATE_CASE==	BUILD STATUS
TOW (SPECIFIC)	A	DROP LOCATION	Lat/long as shared by the customer on the Location Page	DropLocationLat / DropLocationLong	CAPTURED AND SAVED IN CREATE TICKET LOCATION PAGE
	B	VENDOR REACH DROP LOCATION	Lat/long captured at the click of REACH (on-the-way-to-drop page)	`RSP_Drop_Latitude` / `RSP_Drop_Longitude`	CAPTURED AND SAVED on the vendor app Reached Drop action. Your version reads NOT CAPTURED — this was built on 11 September.
COMMON (RSR & TOW)	C	BREAKDOWN LOCATION	Lat/long as shared by the customer on the Location Page	Latitude / Longitude	CAPTURED AND SAVED IN CREATE TICKET LOCATION PAGE
	D	VENDOR START LOCATION	Lat/long captured at the click of ACCEPT	RSP_Start_Latitude / RSP_Start_Longitude	CAPTURED ON VENDOR APP "ACCEPT" PAGE
	E	VENDOR REACH BDL LOCATION	Lat/long captured at the click of REACH (on-the-way page)	`Reach_Location_Lat` / `Reach_Location_Lon`	CAPTURED AND SAVED on the vendor app Reach action, and already used by the closure calculations. Your version reads NOT CAPTURED — this was built on 11 September.
	F	VENDOR CANCEL LOCATION	Lat/long captured at the click of CANCEL (on-the-way page)	Cancel_Location_Lat / Cancel_Location_Lon	CAPTURED ON VENDOR APP REACH BREAKDOWN LOCATION PAGE
	G	VENDOR FLEET LOCATION	Nearest fleet location, looked up from the	Vendor_Fleet_Location_Lat / _Lon	LOOKED UP FROM THE VENDOR TABLE — nearest of Base_Location_1..5_Lat/_Lon, chosen

SERVICE TYPE	VAR	LOCATION HEAD	SOURCE	==FIELD(S) ON CREATE_CASE==	BUILD STATUS
			vendor table		by straight-line comparison
	H	RESCUE OFFICE LOCATION	STATIC [12.967945122837245, 77.6110507612277]	Constant in all four applications	STATIC LOCATION — matches your coordinate exactly, to all 15 decimal places

1. At vendor acceptance

Trigger. ETA_To_Breakdown_repair, Distance_To_Breakdown_repair, ETA_To_Breakdown_TOW and Distance_To_Breakdown_TOW are written when the vendor taps 'Accept' in their app.

ETA_TO_DROP_TOW and Distance_TO_DROP_TOW are written when the vendor taps 'Picked-Up' in their app. Built and live since 11 September — your email records this as omitted entirely. It is stored under different field names, shown in the table below.

Measured from: the vendor's live GPS position at that moment → the breakdown location. For the picked-up pair the system uses the vendor's live position at the Picked-Up tap, falling back to the breakdown location on the ticket. Your document specifies the stored REACH BREAKDOWN position instead. In practice the vendor is standing at the vehicle at that moment, so the two are metres apart — confirm if you want it switched.

FIELD IN YOUR DOCUMENT	==FIELD IN THE SYSTEM==	VENDOR ACCEPTS	AGENT ACCEPTS ON VENDOR BEHALF	SERVICE / TRIGGER
ETA_To_Breakdown_repair	`ETA`	Travel time from the vendor accept location to the breakdown location (mins)	INPUT BY AGENT — the field unlocks for the agent when Acceptance Mode is agent-on-behalf	Repair — vendor clicks "accept"
Distance_To_Breakdown_repair	`Distance_To_Breakdown`	Road distance from the vendor accept location to the breakdown location (km)	INPUT BY AGENT	Repair — vendor clicks "accept"
ETA_To_Breakdown_TOW	`ETA_TOW`	Travel time from the vendor accept location to the breakdown location (mins)	INPUT BY AGENT	Tow — vendor clicks "accept"
Distance_To_Breakdown_TOW	Distance_To_Breakdown_TOW — name matches	Road distance from the vendor accept location to the breakdown location (km)	INPUT BY AGENT	Tow — vendor clicks "accept"
ETA_TO_DROP_TOW	`Return_Journey_ETA`	Travel time from the pickup position to the drop location (mins)	INPUT BY AGENT	Tow — vendor clicks "picked-up"

FIELD IN YOUR DOCUMENT	==FIELD IN THE SYSTEM==	VENDOR ACCEPTS	AGENT ACCEPTS ON VENDOR BEHALF	SERVICE / TRIGGER
Distance_TO_DROP_TOW	Breakdown_To_Drop_Distance	Road distance from the pickup position to the drop location (km)	INPUT BY AGENT	Tow — vendor clicks “picked-up”

Only the pair matching the ticket’s service type is written. These are the figures shown to the customer and used to judge ETA Met / Missed.

On the naming differences: we would prefer to keep the live field names and work from this mapping rather than rename them. These fields already hold production data on every closed case, and renaming a Zoho field breaks every report, filter and export that points at it. The mapping costs nothing and carries no risk — please confirm this is acceptable.

2. At agent closure

Trigger. Vendor_Distance and Vendor_Round_Trip_Distance are calculated when the agent saves the Agent Closure step. This is the only point they are calculated — an unclosed case leaves them empty.

`Roundtrip\_Distance` is now calculated when the ticket is confirmed, at the time Task Confirmation Time is stamped — built 8 October, as you asked. The Agent Closure calculation is kept as a fallback: it only ever fills the field when it is empty, so it can never overwrite the earlier value. This matters for a tow case confirmed before its drop location is known, where the earlier calculation has nothing to measure and deliberately writes nothing.

FIELD NAME	ROUTE MEASURED	TRIGGER POINT	APPLIES TO	==BUILD STATUS==
Vendor_Distance	1. If vendor reach location is not null: vendor start → vendor reach 2. Else if vendor cancel location is not null: = Cancel Distance (from section 3) 3. Else: vendor base location → breakdown location [agent entry]	Agent saves the Agent Closure step	All REPAIR tickets	1. BUILT. 2. BUILT 8 October — it reuses the `Cancel_Distance` the vendor app already recorded, so no extra Google call is made. 3. NOT BUILT — decision needed, see below.
Vendor_Round_Trip_Distance	1. If vendor reach location and vendor drop location are both not null: base → vendor reach → vendor drop → base 2. Else if vendor reach location is not null and vendor drop location is null: base → vendor reach → base 3. Else if vendor cancel location is not null: base → vendor cancel point → base 4. Else if agent closure status is cancelled- billable or cancelled-not-billable: base → pickup [agent entry] → base 5. Else: base → pickup [agent entry] → drop [agent entry] → base	Agent saves the Agent Closure step	All TOW tickets	1. BUILT. 2. BUILT 8 October — but placed last rather than second, see the note below. 3. BUILT. 4. PARTLY BUILT — only cancelled- billable is handled; decision needed. 5. BUILT.

FIELD NAME	ROUTE MEASURED	TRIGGER POINT	APPLIES TO	==BUILD STATUS==
Roundtrip_Distance	1. Office → breakdown location [agent entry] → Office	Time of Task confirmation	All REPAIR tickets	BUILT 8 October at the new trigger.
Roundtrip_Distance	1. Office → pickup [agent entry] → drop [agent entry] → Office	Time of Task confirmation	All TOW tickets	BUILT 8 October at the new trigger. If the drop location is not yet set at confirmation time, nothing is written and Agent Closure fills it instead.

On the ordering of vendor round-trip branch 2: your list places it before the cancel branch. We placed it after, so that it can only ADD a figure to cases that produce none today and can never change one that is already being paid. In practice the two routes differ by metres — a vendor cancels where they reached. Moving it to the literal position is a one-line change if you want it; we did not want to alter existing cancelled-billable payments without asking.

- Vendor_Distance and Vendor_Round_Trip_Distance are mutually exclusive — repair writes the first, tow writes the second.
- **Vendor_Distance is the figure that pays the REPAIR vendor** (it feeds the KM fee).
- **Vendor_Round_Trip_Distance is the figure that pays the TOW vendor** (it feeds the KM fee).
- Roundtrip_Distance is office-based, not vendor-based — it will be much larger on the same case.

Both figures are read in four places: the Final Closure screen, the vendor fee calculation, the Vendor Accounting screen and the accounting export. On the Vendor Accounting screen, a repair case whose `Vendor\_Distance` is missing falls back to `Distance\_To\_Breakdown` so it is never priced at 0 km — added 6 October after you reported exactly that. The tow figure is editable at Operations Approval and recalculates the fee live.

3. At cancellation

Trigger: the vendor cancels in their app, measured from their GPS at that moment.

FIELD	ROUTE MEASURED	==BUILD STATUS==
Cancel_Distance	Vendor start → where they cancelled	BUILT
Distance_Cancel_To_Breakdown	Cancel point → breakdown location	Already haversine — this calculation has never used the Google API, so your change required no work.

Not written when an agent cancels on the vendor's behalf — there is no vendor GPS position to measure from. Implemented: both fields are hidden and left unwritten whenever Acceptance Mode is agent-on-behalf.

4. Not calculated by Google

FIELD	SOURCE	==BUILD STATUS==
Office_To_Mechanic	Typed by agent (agent-on-behalf only)	DOES NOT MATCH THE BUILD. The vendor applications calculate and write this at work-completed, for every case. Please confirm which behaviour you want.
Breakdown_To_Office	Typed by agent (agent-on-behalf only)	DOES NOT MATCH THE BUILD — same as above, calculated and written rather than typed.
Agent_ETA_Override	Typed by agent to replace the vendor's ETA	BUILT
ETA Met / Missed	Calculated in-app: arrival time vs ETA	BUILT
Vendor_Fleet_Location_Lat / _Lon	Nearest vendor base, picked by straight-line comparison	BUILT

5. Five things

#	POINT	==BUILD STATUS==
1	Calculated once, only when empty. An existing value — measured or typed — is never overwritten. Re-saving a closure recalculates nothing.	BUILT
2	Straight-line fallback. If Google cannot route a leg (no network, API limit, unroutable point), straight-line distance is used instead of leaving the field blank. That reads 20–40% low.	BUILT
3	What leaves a field blank. Closure calculations need coordinates on the ticket, an identifiable vendor, and that vendor having a base location on file. If any is missing the calculation is skipped.	BUILT — and each skip now records which of the three was missing, so a 0 km travel distance can be traced to its cause rather than guessed at.
4	The ETA displayed to the customer on WhatsApp must be in TIME FORMAT (e.g. 15:35). Calculated ETA + 10 minutes for cases run by vendors on their app. For cases run by an agent on behalf of a vendor, ACCEPTANCE TIME + Agent ETA Override.	BUILT 8 October in all four applications, both rules, 24-hour format. NOT YET LIVE, and it needs an action on your side: the approved WhatsApp template `share_rescue_service_details_and_eta` has the literal word “Hrs” written into its body after the ETA variable, so a clock time would reach the customer as “15:35 Hrs”. The template text must be edited in WhatsPortal and re-approved by Meta. The code switches over the moment it is approved. Please confirm who makes that change.
5	All TOW cases calculate distance in 4-wheeler mode. All REPAIR cases calculate distance in 2-wheeler mode.	Already in place — tow routes as DRIVING, repair as TWO_WHEELER. No work was required.

6. Decisions we need from you

Added to your document. Each of these is held open deliberately rather than decided for you, because each one changes what a vendor is paid or what a customer is told.

#	DECISION	WHY WE HAVE NOT DECIDED IT
1	`Vendor_Distance` branch 3 — should a repair case cancelled before the vendor left base still generate a KM fee?	As written, branch 3 applies to all repair tickets with no status condition. Taken literally it pays a kilometre fee on a case where the vendor never travelled. That is a billing decision, not a technical one.
2	Vendor round-trip branch 4 — should cancelled-not-billable pay a KM fee, as the wording implies?	Only cancelled-billable is handled today. Extending it means paying on non-billable cancellations.
3	Vendor round-trip branch 2 ordering — keep it last (adds a figure only), or move it to the literal position in your list?	The literal position changes which formula existing cancelled-billable cases use. We did not want to alter payments already being made without asking.
4	Field naming — confirm the mapping in section 1 is acceptable rather than renaming live fields.	Renaming a Zoho field that holds production data breaks every report, filter and export pointing at it.
5	`Office_To_Mechanic` / `Breakdown_To_Office` — typed by the agent, or calculated as they are today?	Your section 4 and the build disagree. Either is straightforward; we need to know which is intended.
6	The breakdown-to-drop origin — measure from the stored vendor Reach position rather than their live position at Picked-Up?	The two are metres apart in practice. Changing it is simple but it alters a figure already being recorded.
7	The WhatsApp template text — who edits it and re-submits it to Meta?	Section 5 point 4 cannot go live until this is done.
8	Which document governs where your email and this attachment differ?	Your email asks that <code>Vendor_Distance</code> be removed as unaligned, but this document keeps it and names it the REPAIR payment figure. Your email also asks that Rescue Round-Trip use the vendor's click locations, but both <code>Roundtrip_Distance</code> rows here specify [agent entry] locations — which is what is built. We have followed this document; please confirm that is right.